
Steer-to-Detect: Probing Hidden Representations for Detection of LLM-Generated Texts

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Abstract

The rapid advancement of large language models (LLMs) has made machine-generated text increasingly difficult to distinguish from human-written text. While recent studies explore leveraging internal representations of language models to uncover deeper detection signals, these raw features often exhibit substantial overlap between classes, limiting their discriminative power. To address this challenge, we propose Steer-to-Detect (S2D), a two-stage framework for detecting LLM-generated text. In the first stage, S2D learns a steering vector that is injected into the hidden states of a frozen observer LLM, producing representations with improved class separability. In the second stage, detection is performed via a hypothesis testing procedure based on the steered representations. We establish finite-sample, high-probability guarantees for Type I and Type II errors, providing a theoretical characterization of the procedure. Empirically, S2D achieves strong and consistent performance across a range of settings, including out-of-distribution scenarios and adversarial perturbations.

1 Introduction

Large language models (LLMs) are increasingly deployed across a broad range of domains, including education, finance, legal services, customer support, and writing assistance [1–5]. Despite their widespread adoption, their use has raised growing societal concerns, including the spread of misinformation, academic misconduct, and the erosion of trust in written content [6–9]. A key reason is that LLM-generated text often closely resembles human writing, making it difficult to distinguish between the two. This challenge highlights the need for reliable detection of LLM-generated text.

Early work on this problem suggests that as long as distributional differences exist, reliable detection is possible in principle, particularly with sufficiently long samples [10]. Building on this observation, a prominent line of work seeks to explicitly introduce such differences during generation, most notably through watermarking methods that embed structured patterns identifiable with statistical guarantees [11, 12]. While effective under controlled settings, these approaches rely on access to the generation process and are not applicable when watermarking is absent or not widely adopted. This limitation motivates passive methods, which aim to infer the origin of a text directly from observed samples without any control over the generation process. A detailed taxonomy is provided in Section 1.1.

A key challenge in the passive setting is that the distributional differences between human-written and LLM-generated text can be subtle, particularly for short or moderate-length samples. This raises a natural question: rather than relying solely on these weak signals, can we amplify them to improve detectability? Recent studies suggest that hidden representations of LLMs encode behavior-relevant information that is not fully captured by final-layer outputs [13–17]. These hidden representations provide a richer feature space in which the differences between human- and machine-generated text may become more pronounced. This observation motivates our central question: *Can we*

reshape LLM hidden representations to amplify such differences and enable reliable detection of machine-generated text?

Our Contributions. In this paper, we propose *Steer-to-Detect* (S2D), a novel method for detecting LLM-generated text. At a high level, S2D employs a surrogate model as an “observer” and intervenes in its representation extraction process. Specifically, during the forward pass on input text, we steer hidden representations by adding a universal vector, thereby modifying how representations are formed. The vector is learned to enhance the separability between human-written and LLM-generated text. This approach avoids expensive model fitting and extends to other binary detection tasks.

On the theoretical side, we establish finite-sample control of the Type I (or false positive) error and derive an explicit upper bound on the excess Type II (or false negative) error. In addition, we provide a quantitative characterization of robustness, demonstrating the reliability of the detector in risk-sensitive settings.

Empirically, we conduct extensive experiments and show that S2D achieves strong and consistent performance across both in-distribution (ID) and out-of-distribution (OOD) settings. Further analyses demonstrate robustness to paraphrasing, adversarial perturbations, and variations in input length. In short, we propose a simple yet effective approach that enhances detectability by steering hidden representations.

1.1 Related Works

LLM-generated Text Detection. Active methods modify the generation process and are beyond the scope of this work [29–31], so we focus on passive detection. As summarized in Table 1, passive detectors can be organized along two axes: whether labeled training data are required and whether detection relies on auxiliary rewritten or perturbed variants of the input text. This yields four broad categories: train-free rewrite-based [10, 18], train-free non-rewrite-based [19–21], train-based rewrite-based [22, 24], and train-based non-rewrite-based methods [25–28]. Rewrite-based methods incur additional cost due to auxiliary text generation, whereas non-rewrite-based methods avoid this overhead. Our method falls into the *train-based, non-rewrite-based* category. Prior work in this setting typically relies on final-layer outputs such as logits or derived scores, leaving intermediate representations underexplored [16]. More recent studies have begun to use hidden representations for detection [32, 33], with RepreGuard [33] showing that hidden representations can provide stronger signals than final-layer outputs. Unlike these methods, we do not treat representations as fixed features. Instead, we shape their geometry to better separate human-written and LLM-generated text.

Table 1: Taxonomy of passive detection.

	Rewrite	Non-rewrite
Train-free	[10, 18]	[19–21]
Train-based	[22–24]	Ours & [25–28]

Representation Engineering. Representation engineering studies how information encoded in the hidden states of LLMs can be analyzed, interpreted, and manipulated to understand, monitor, or control model behavior [34, 13, 35, 16, 36–42]. This line of work includes both methods for characterizing what hidden representations encode and intervention-based approaches that modify them. Among these, *activation steering* has emerged as a prominent technique that perturbs intermediate activations along task-specific directions (often represented as steering vectors) in representation space, enabling targeted behavioral changes without updating model parameters [43, 44]. Recent studies apply activation steering to improve factual reliability, control generation style, and monitor model behaviors such as hallucinations and toxic content using internal representations [45–49, 41]. Despite these advances, the use of representation-level interventions for LLM-generated text detection remains relatively underexplored. In this work, we repurpose activation steering to reshape the geometry of hidden representations, transforming raw hidden states into features that better separate human-written and LLM-generated text.

2 Our Method: Steer-to-Detect (S2D)

Problem Formulation. Let $\mathcal{X}$ denote the space of all text sequences (possibly of varying length). We assume that human-written and LLM-generated texts are drawn from two distributions, $\mathbb{P}_0$ and $\mathbb{P}_1$, respectively. Given an observation $x \in \mathcal{X}$, detection is formulated as the binary hypothesis test:

$$\mathcal{H}_0 : x \sim \mathbb{P}_0 \quad (\text{human-written}) \quad \text{versus} \quad \mathcal{H}_1 : x \sim \mathbb{P}_1 \quad (\text{LLM-generated}). \quad (1)$$

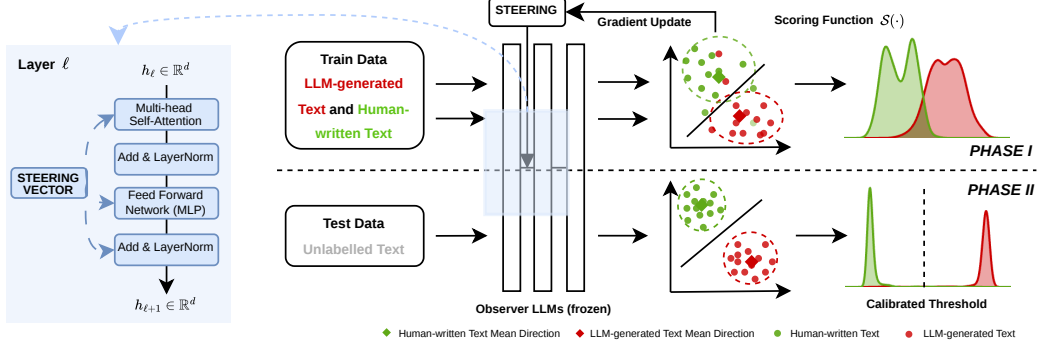


Figure 1: **Overview of Steer-to-Detect (S2D).** *Phase I* (top row) applies a steering vector to reshape the observer LLM’s hidden representations, enhancing the separation between human-written and LLM-generated text. *Phase II* (bottom row) scores unseen texts and rejects the null hypothesis when the score exceeds a calibrated threshold.

2.1 Method Overview

We begin with an overview of our method. We employ a surrogate model as an “observer” and extract hidden representations during the forward pass on input text. Representations of LLM-generated and human-written text exhibit systematic differences, which, although subtle, provide a sufficient signal for detection [33]. Motivated by this observation, we intervene in the representation formation process by injecting a lightweight, learnable steering vector $\mathbf{v}$ into intermediate hidden states during the forward pass. This modifies how representations are formed and enhances their separability between the two types of text. Based on the resulting representations, we construct a hypothesis test for detection. The overall framework is illustrated in Figure 1.

More specifically, let $f_\theta : \mathcal{X} \rightarrow \mathbb{S}^{d-1}$ denote the representation map of an observer model, which embeds a text sequence into a d -dimensional vector on the unit hypersphere (to be specified later). To enable intervention in the representation space, we introduce a steering vector $\mathbf{v} \in \mathbb{R}^d$ and denote the resulting steered representation map by $f_{\theta, \mathbf{v}}$.

Our method proceeds in two stages, as introduced earlier. In the first stage, we estimate the steering vector $\mathbf{v}$ from training data $\mathcal{S}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{n_1}$ and fit parametric class-conditional models to the steered representations. In the second stage, we construct a likelihood-ratio statistic based on the estimated models and use it to test whether a given text is human-written or LLM-generated. We next describe (i) the computation of f_θ and the introduction of $\mathbf{v}$ in Section 2.2, (ii) the learning of $\mathbf{v}$ in Section 2.3, and (iii) the detection procedure in Section 2.4.

2.2 Representation Extraction and Steering

Representation Extraction. We define the mapping f_θ by extracting hidden states from an observer model with L Transformer blocks and hidden dimension d . For a sequence $x \in \mathcal{X}$, let $h_{\ell, t}(x) \in \mathbb{R}^d$ denote the hidden state at layer $\ell (\leq L)$ and token index t . Prior work [50, 13, 16, 33] suggests that tokens near the end of a sequence encode richer contextual information and exhibit more pronounced differences between human-written and LLM-generated text, as later tokens aggregate broader preceding context in causal LLMs. Motivated by this observation, we focus on the final $K \in (0, 1]$ fraction of valid (non-padding) tokens,¹ indexed by $\mathcal{I}_K(x)$. In addition, intermediate layers capture complementary information beyond the final layer [16], so we aggregate hidden states from the last N layers to obtain a more stable sequence-level representation. Mathematically, we define

$$f_\theta(x) = \frac{\bar{m}(x)}{\|\bar{m}(x)\|} \in \mathbb{S}^{d-1} \quad \text{with} \quad \bar{m}(x) := \frac{1}{N \cdot |\mathcal{I}_K(x)|} \sum_{\ell=L-N+1}^L \sum_{t \in \mathcal{I}_K(x)} h_{\ell, t}(x) \in \mathbb{R}^d,$$

where $|\mathcal{I}_K(x)|$ denotes the number of selected non-padding tokens.

Remark 2.1 (Representation Norms and Directionality). *In Appendix E, we provide empirical support for this modeling choice f_θ . In modern Transformer models, representation norms are often*

¹Non-padding tokens refer to tokens corresponding to actual text content, excluding padding tokens.